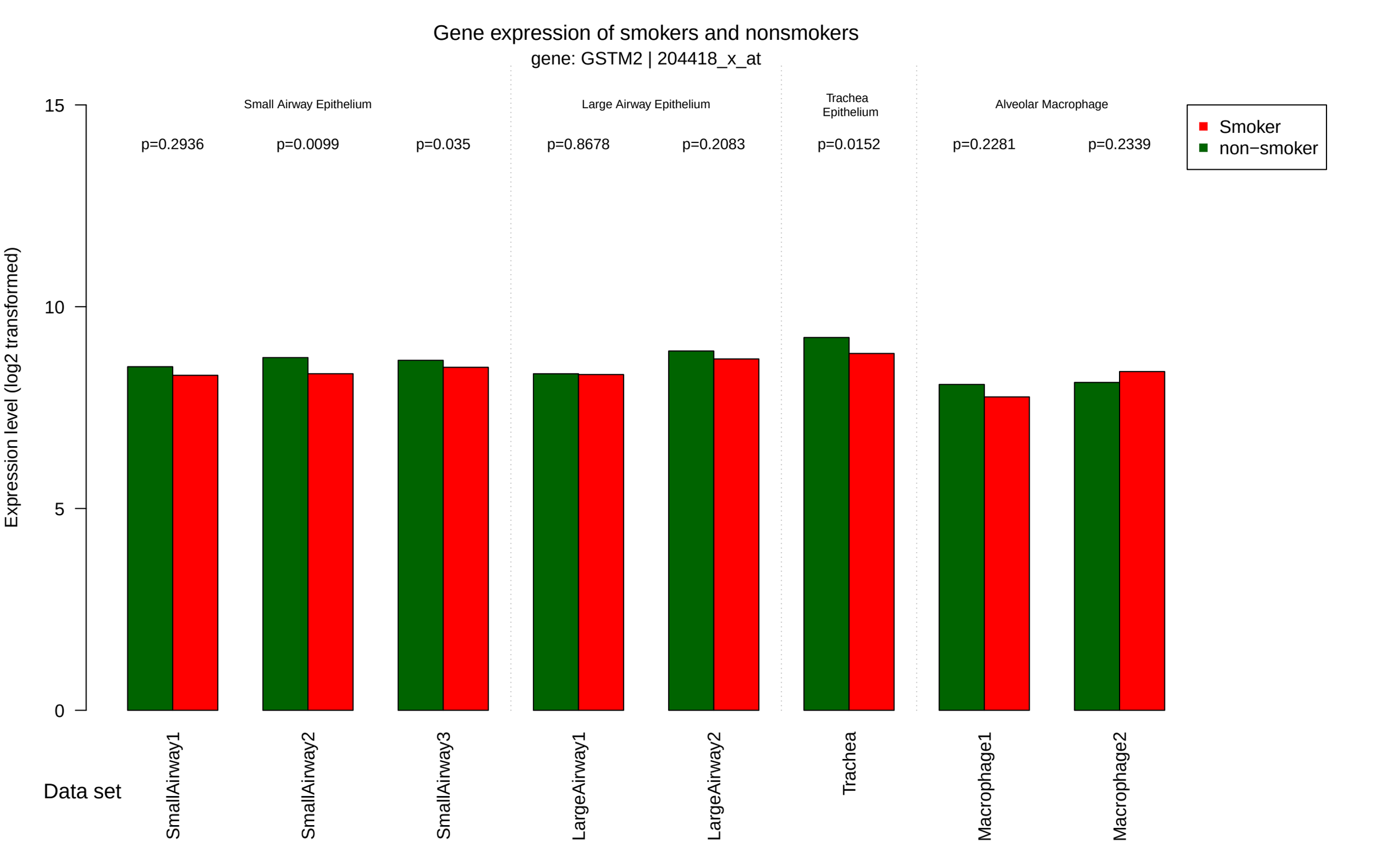
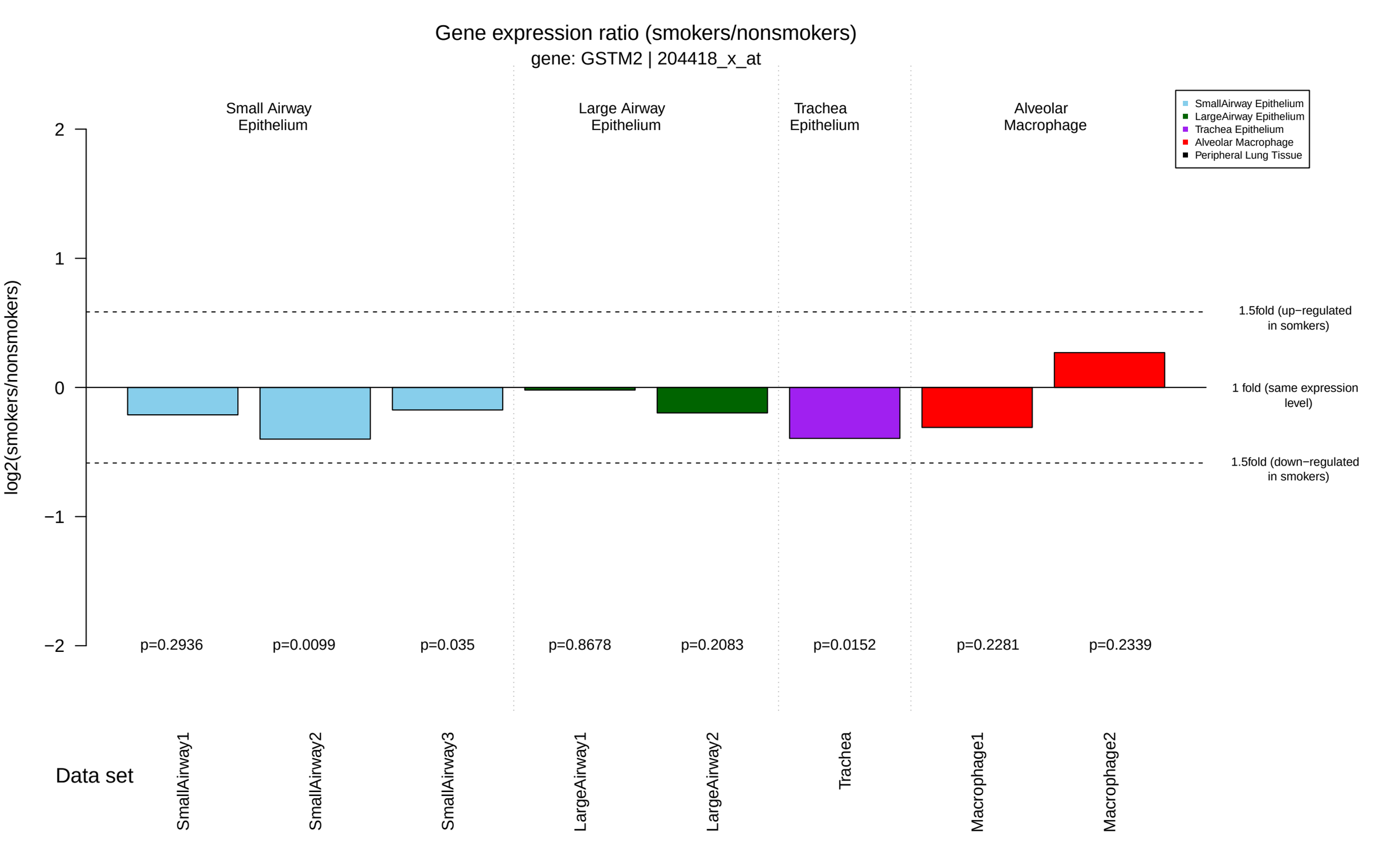
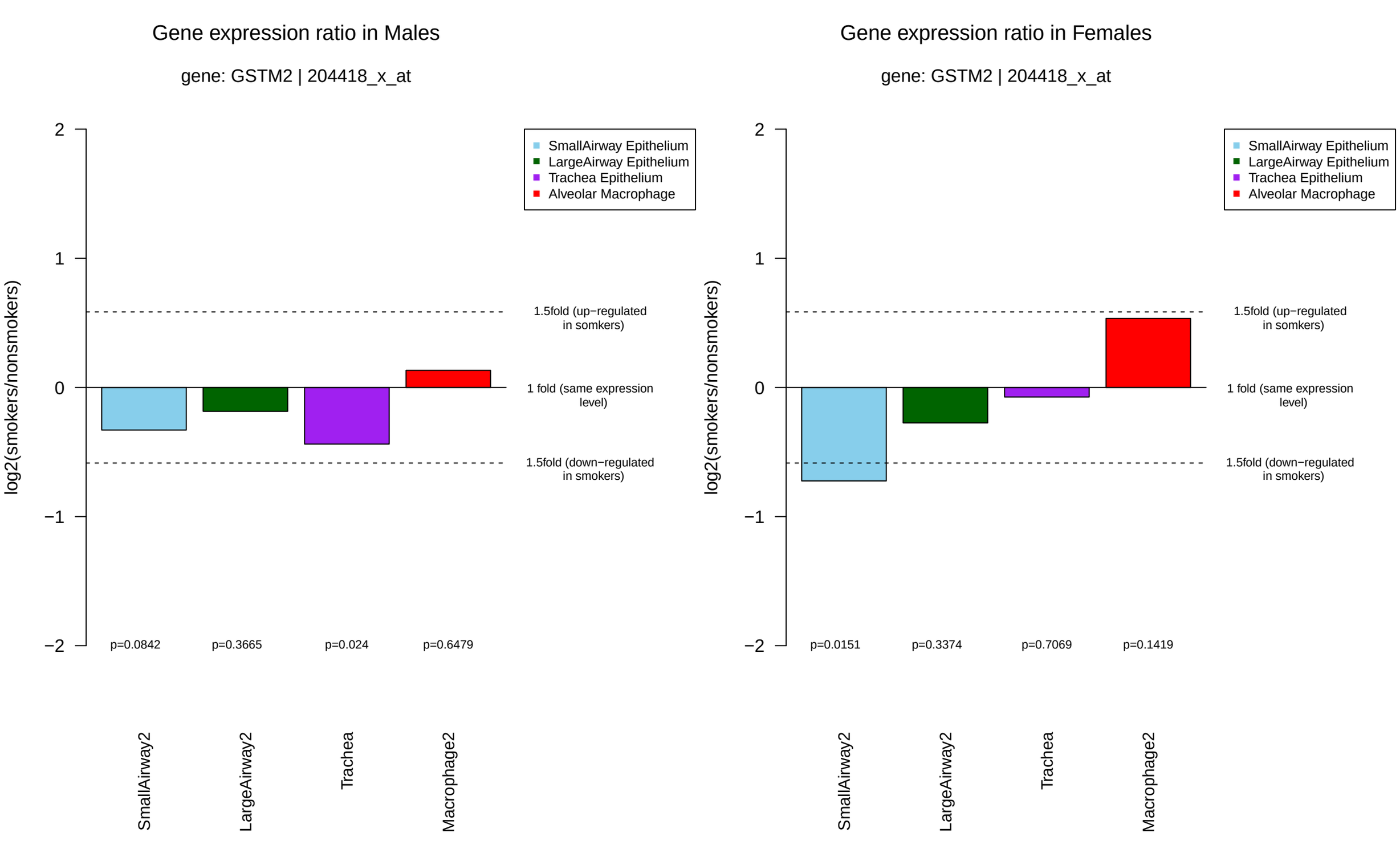
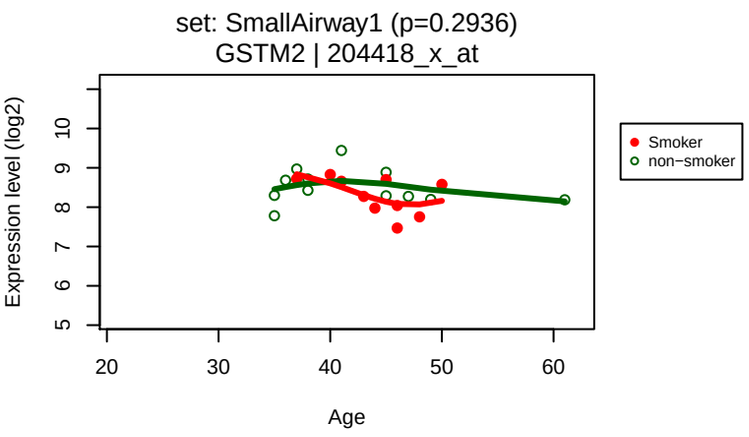
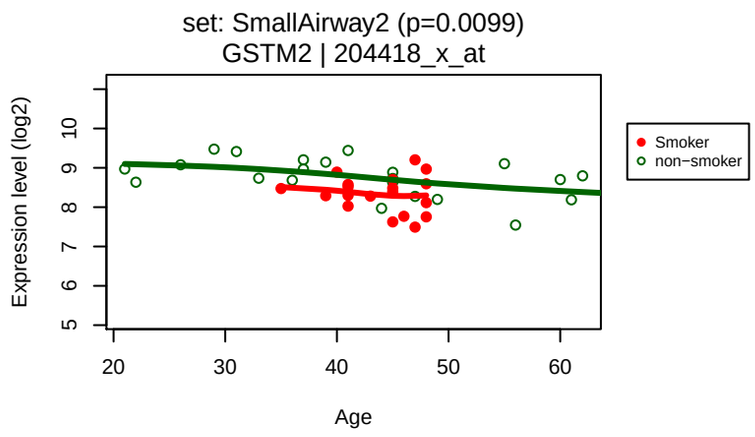
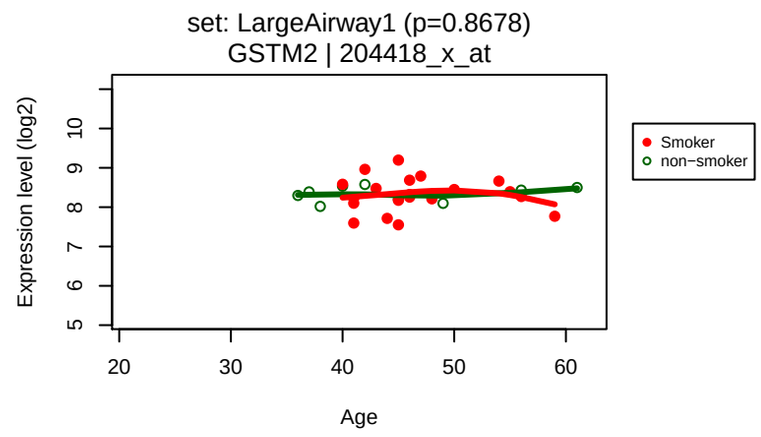

Gene expression of smokers and nonsmokers
gene: GSTM2 | 204418_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2936
p=0.0099
p=0.035
p=0.8678
p=0.2083
p=0.0152
p=0.2281
p=0.2339
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

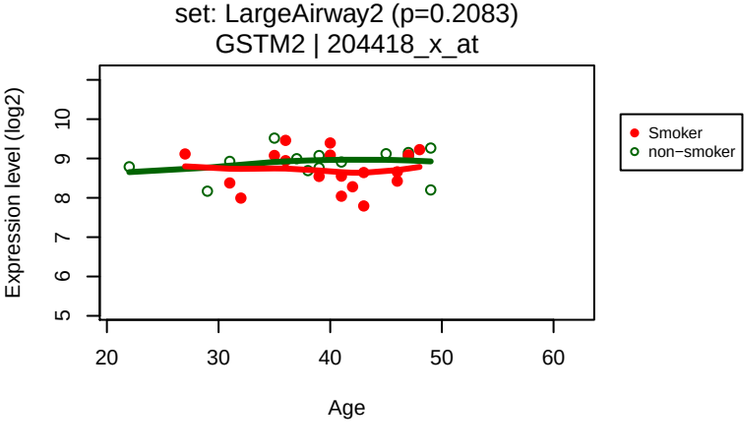

Gene expression ratio (smokers/nonsmokers)
gene: GSTM2 | 204418_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2936
p=0.0099
p=0.035
p=0.8678
p=0.2083
p=0.0152
p=0.2281
p=0.2339
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

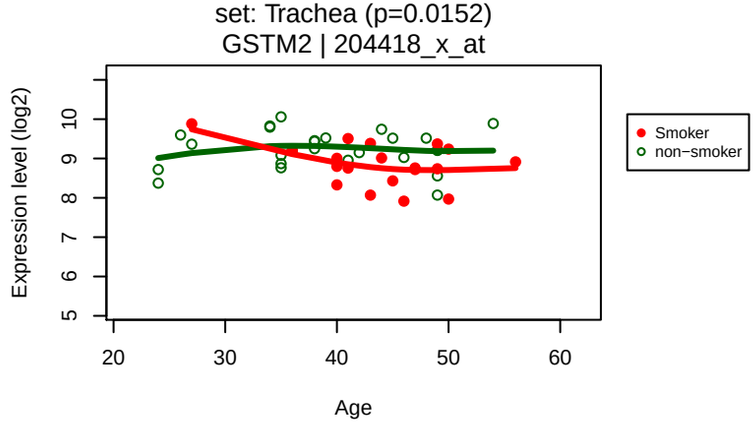

Gene expression ratio in Males
gene: GSTM2 | 204418_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0842
p=0.3665
p=0.024
p=0.6479
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSTM2 | 204418_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0151
p=0.3374
p=0.7069
p=0.1419
SmallAirway2
LargeAirway2
Trachea
Macrophage2

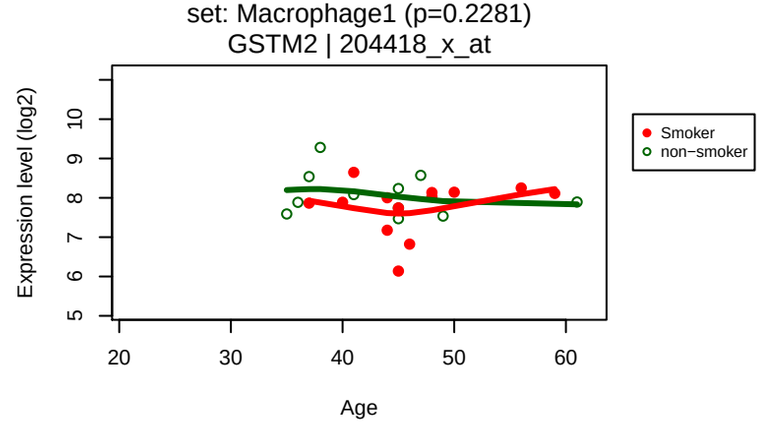

set: SmallAirway1 (p=0.2936)
GSTM2 | 204418_x_at
Expression level (log2)
Age
Smoker
non−smoker

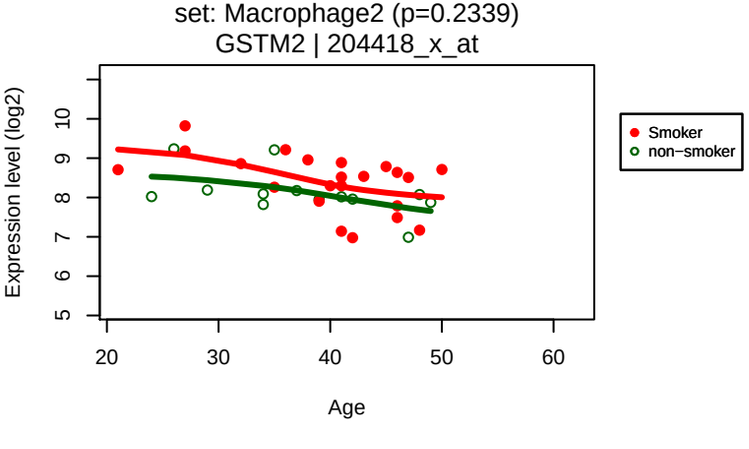

set: SmallAirway2 (p=0.0099)
GSTM2 | 204418_x_at
Expression level (log2)
Age
Smoker
non−smoker

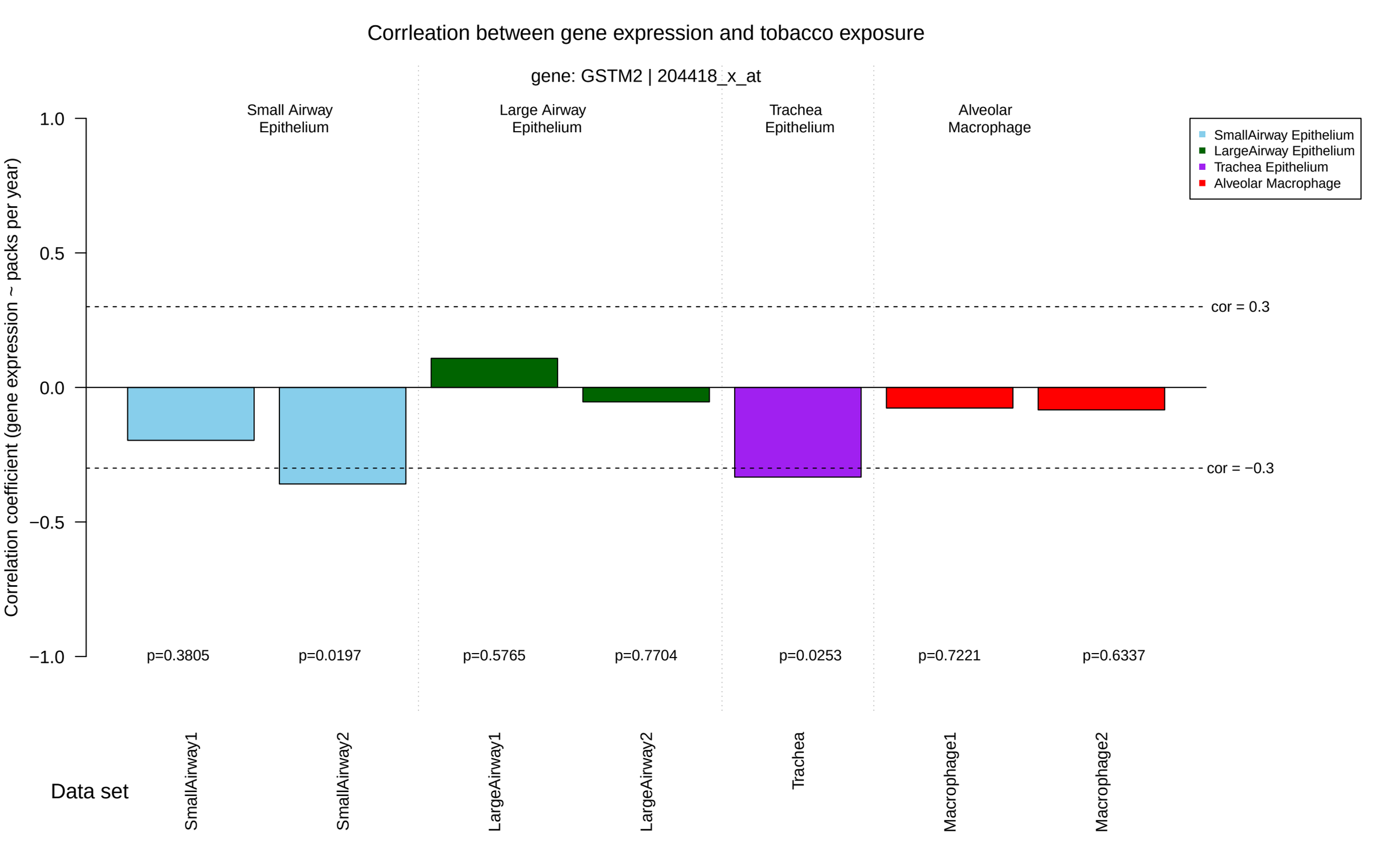

set: LargeAirway1 (p=0.8678)
GSTM2 | 204418_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2083)
GSTM2 | 204418_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0152)
GSTM2 | 204418_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2281)
GSTM2 | 204418_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2339)
GSTM2 | 204418_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSTM2 | 204418_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3805
p=0.0197
p=0.5765
p=0.7704
p=0.0253
p=0.7221
p=0.6337
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2